

Pediatric Patient Counting with Interactive Object Detection

An Exploratory Project in Hospital Data Collection

Peter Amoah Mensah Fafali Dorkunor

April 25, 2026

The Problem: Data Invisibility

- **The Blind Spot:** Hospital statistics frequently underestimate the true pediatric load because they only record registered patients, completely missing accompanying children.[1]
- **Context:** In many LMIC hospitals, women commonly carry babies or toddlers strapped to their backs through outpatient departments.[1]
- **Operational Impact:** Without a precise count of children on the premises, administrators struggle to allocate pediatric triage resources, design safe waiting areas, and plan vaccination campaigns.[1]

Our Solution: Multi-Metric Differentiation

- **Automated Counting:** We propose a camera-based system at hospital entrances that counts total entrants and increments a child counter based on geometric and anatomical thresholds.[1]
- **Primary Metric (Height):** The system initially flags individuals under a calibrated height threshold (e.g., 1.2 meters) as potential children.[1]
- **Data Dashboard:** The counts will be aggregated in a simple database and displayed on a dashboard (e.g., "Hourly counts of children") to provide administrators with actionable demographic data.[1]

The Challenge: Absolute height thresholds fail when encountering short adults or children experiencing growth spurts.

- **Head-to-Body Ratio (HBR):** Human proportions scale systematically with age. We use pose estimation to extract the ratio of head length to total body height.
- **Mathematical Calibration:** A toddler typically presents an HBR of approximately 1:5.6, whereas a mature adult's ratio approaches 1:7.5 or 1:8.
- **Multi-Metric Fusion:** If a detection falls below the height threshold but presents an adult HBR of 1:8 (or mature facial geometry), the system overrides the height metric and classifies the individual as an adult, preventing false positives.

The Challenge: Detecting Carried Children

The Edge Case: A standard height or ratio metric will entirely miss an infant carried on an adult's back.

- **Geometric Indicators:** To tackle this, our algorithm looks for structural anomalies on a single tracked adult bounding box.[1]
- **Pose Estimation Logic:** By analyzing the coordinate structure of the adult's back, the system searches for secondary head structures or a small protrusion of limbs (e.g., coordinates indicating child's legs wrapped around the waist).[1]
- **Inference:** If these geometric conditions are satisfied, we mathematically infer and count a carried child.[1]

The Challenge: Preventing double-counting if a child wanders out of frame and returns.

- **Centroid Tracking:** As a person moves, we calculate the x and y coordinates of their bounding box centroid.
- **Distance Metrics:** We use simple Euclidean distance calculations between centroids across consecutive frames to maintain a persistent ID.
- **Re-entry Thresholds:** If an ID is lost, we hold it in memory for a specified time window (e.g., 60 seconds). If a new detection appears at the boundary with mathematically similar bounding box dimensions and anatomical ratios within that window, we reassign the original ID.

Q: How do we ensure these geometric thresholds are accurate for our specific population? *Answer:* We will conduct a rigorous testing phase, comparing the algorithm's automated counts against manual, ground-truth observations at a busy hospital entrance. This allows us to mathematically fine-tune the HBR weights and height cutoffs to minimize miscounting outliers.[1]

Q: How does this project handle strict hospital privacy rules? *Answer:* The video feed processing will be done entirely locally. No images or identifiable video frames will be stored; the system outputs only numeric counts, ratios, and timestamps.[1]